

Prerak Singh Tanwer

9352492232 | preraktanwer@gmail.com | linkedin.com/in/prerak-tanwer | github.com/Crashz99

Portfolio: prerak-tanwer.vercel.app

EDUCATION

Vellore Institute of Technology

B.Tech Computer Science – Specialisation in Information Security | CGPA: 8.15

Jayendra Saraswati Vidhyalya

Class XII (CBSE) | 85.2%

Air Force School

Class X (CBSE) | 95.1%

Vellore, India

Aug. 2024 – Jun. 2028

Coimbatore, India

Mar. 2024

Coimbatore, India

Mar. 2022

EXPERIENCE

Data Science Engineer

Remote

NexaVerse (Internship)

Jul. 2026 – Present

- Developing an AI-assisted detection engine for **EvolveSafe**, an AI-powered browser-security product focused on phishing and JavaScript-obfuscation detection, alongside voice AI automation and LLM memory-system initiatives

Summer Intern

Panchkula, Haryana, India

Bharat Electronics Limited (BEL)

May 2026 – Jul. 2026

- Engineered a two-stage **YOLOv8 + InsightFace (SCRFD + ArcFace)** recognition pipeline with temporal-consensus voting and quality-gated auto-enrollment, achieving near-perfect accuracy on multi-angle validation testing and solving the cold-start problem for live reception CCTV feeds
- Extended the **YOLOv8/ByteTrack** toolchain for radar-less vehicle speed estimation, traffic-density heatmaps, and pedestrian re-identification on a **4x NVIDIA Tesla T4 GPU server**

Software Engineer Intern

India

Indian Army

May 2026 – Jul. 2026

- Designed and developed an internal data-automation application using **HTML, CSS, JavaScript, and SQL**, supporting structured data processing and workflow automation

PROJECTS

AtlasOps | *Next.js, React, TypeScript, Leaflet, Vercel* | [Live Demo](#)

- Built and deployed a configurable operations and data-collection platform with dynamic forms, structured records, role-aware workflows, online mapping, analytics, and organisation-level customisation

Privacy Shield | *Node.js, Chrome Extension API, Etherscan API, GoPlus API* | [GitHub](#)

- Built a weighted heuristic engine to detect malicious Ethereum wallets using real-time security metadata from Etherscan and GoPlus APIs with parallel asynchronous fetching
- Developed a Chrome extension interface for visualising wallet risk scores, transaction velocity, wallet age, and supporting security indicators

Hybrid Plagiarism Detection System | *Python, Flask, Next.js, React, Sentence-Transformers* | [GitHub](#)

- Combined KMP exact string matching with sentence embeddings and cosine similarity to detect both verbatim copying and semantically paraphrased content

VIBECADE | *React, Vite, Web Audio API, IndexedDB, Vercel* | [Live Demo](#)

- Built a local-first interactive focus and music workspace with persistent playlists, Pomodoro and study modes, session history, reactive visual/audio environments, scheduling, mini-games, and browser-local persistence

LEADERSHIP

Manager – Esports, Technosports & Creative Events Committee

Vellore, India

GRAVITAS 2026 – Vellore Institute of Technology

2026

- Managing operations across **22 events** spanning Esports, Technosports, and Creative tracks for GRAVITAS 2026, VIT's flagship techno-cultural fest drawing **20,000** participants annually

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Next.js, Node.js, Vite, Flask, Tailwind CSS, Leaflet, YOLOv8, InsightFace (ArcFace/SCRFD), ByteTrack

Developer Tools & APIs: Git, Vercel, Chrome Extension API, Etherscan API, GoPlus API, Web Audio API, IndexedDB

Spoken Languages: English (Professional), German – Intermediate (A2), Japanese – Elementary (N5), Hindi (Native)

ACHIEVEMENTS & CERTIFICATIONS

VIT Achievers List (2025–2026) | [GRAVITAS Designathon – 2nd Place \(2024\)](#) [[Prototype](#)] | **RIVIERA Creative Writing 2nd Place** (2026)

Alliance University MUN – **Honourable Mention** | **Best Delegation Award**

Certifications: Mastering DSA – Abdul Bari (Udemy) | Indian Space Academy (ISA) (2025)

Internship Certificates: Bharat Electronics Limited (BEL) | Indian Army

EXTRACURRICULAR

Mozilla Firefox Club – PR & Outreach | **VITMUNSOC** – Member